VEER SURENDRA SAI UNIVERSITY OF TECHNOLOGY (VSSUT), ODISHA

Even Mid Semester Examination for Academic Session 2024-25

COURSE NAME: B.TECH | SEMESTER: 6TH

BRANCH NAME: IT

SUBJECT NAME: SOFTWARE ENGINEERING

FULL MARKS: 30 | TIME: 90 Minutes

Answer All Questions.

The figures in the right hand margin indicate Marks. *Symbols carry usual meaning.*

Q1.		**Answer all Questions.**	[2 × 3]
	a)	Write the features of Software which makes it different from other engineering products (e.g Hardware etc.) and justify why a software does not wear out?	- CO1
	b)	Write the deliverables for each phase of SDLC and distinguish between two types of prototypes used for software projects.	- CO2
	c)	Illustrate the similarities and differences between DFD and ER models.	- CO3
Q2.			[4]
		a) Distinguish between a software and a program. Why a software is not manufactured? Which SDLC phase consumes maximum effort during software development? b) Discuss the spiral and agile software development model with schematic diagrams and advantages, disadvantages.	- CO1
		OR	[4]
		a) How SDLC can be used to deliver high quality of software. State the usefulness of the lifecycle approach over ad-hoc approach for software development. b) Distinguish between different variants of waterfall model with necessity, advantages, disadvantages and schematic diagram.	- CO1
Q3.			[4]
		a) What is the significance of software requirements? How to capture the requirements via elicitation? State the features of a good SRS. b) State the five tasks of software project manager. What is meant by the size of a software project? Why Project manager estimate it? Discuss the size estimation methods also.	- CO2
		OR	[4]
		a) Explain how the project size be estimated by delphi and expert judgement method. Also, state their advantages and disadvantages. b) What is a milestone in software development? Write the advantages to have milestones during software development duration. Justify, how frequently milestones be placed?	- CO2
Q4.			[8]
		a) Discuss the function point metric and LOC metric in detail with associated pros and cons. b) Discuss the COCOMO Model in detail with schematic diagrams for three types of the model.	- CO3
		OR	
		a) What is a risk? Why risk management is necessary? How does risk management affects the overall project cost. b) Define change control and version control with their need. How both are managed using a software management tool?	- CO3

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Total Pages—10

B. Tech-6th(CSC)

Software Engineering

Full Marks : 50

Time : $2\frac{1}{2}$ hours

Answer **all** questions

The figures in the right-hand margin indicate marks

Symbols carry usual meaning

1. Answer *all* questions : 2×5

(*a*) Do you agree with the statement : "Code" is the most important reuse artifact that can be used during software development ? Justify your answer.

(*b*) What do you mean by Structural complexity of a program ? Define a metric for measuring the structural complexity of a program ? How structural complexity of a program different from its computational complexity ?

(*Turn Over*)

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(*c*) What is mutation testing and data flow based testing ?

(*d*) What do you understand by the clean room strategy ? What are its advantages ?

(*e*) What is the purpose of baseline metrics ? Discuss.

2. (*a*) Briefly explain the process of developing software using prototype model. Under what circumstances is it beneficial to develop a prototype model ? Give the advantages and disadvantages of using this model. 4

(*b*) Explain CFG. Draw a neat CFG for the following C function named find maximum. From the CFG determine the Cyclomatic complexity.

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```
int find-maximum(int i, int j, int k){
    int max;
    if(i>j) then
if(i>k) then max=i;
else max=k;
else if(j>k) max=j;
else max=k;
return max;
}
```

4

Or

(*a*) Discuss the laws related to maintenance. The changes made to a software product to add some desired features can be considered as which type of maintenance ? Justify your answer. 4

(*b*) Explain how software cost estimation can be carried out using COCOMO model ? What is the limitation of cost estimation model ? 4

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3. (*a*) Develop the SRS document for the following problem :

Consider the following Automobile Spare Parts Shop Automation Software (AAS) which is required by a retail automobile spare parts shop to automate various book-keeping activities associated with its business.

(*i*) The retail shop deals with a large variety of automobile spare parts procured from various manufacturers. The shop owner maintains different parts in wall mounted and numbered racks.

(*ii*) At the end of each day, the shop owner would request the computer to generate indents for the items which are out of stock. The computer should print out the parts description, the quantity required and the address of the vendor supplying the part. The

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shop owner would have to simply put these printouts into envelopes and courier them to the address printed.

(*iii*) Whenever new supplies arrive, the shop owner should be able to update the inventory.

(*iv*) Whenever any sales occurs, the shop owner would enter the code number of the parts and the corresponding quantities sold. AAS should print out the cash receipt, maintain the cash book and adjust the inventory.

(*v*) The computer should also generate the revenue for any specified day and month, when queried by the owner. 3

(*b*) For the Automobile Spare Parts Shop Automation Software (AAS) given in Q.3(a), draw the following diagrams.

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(*i*) Draw the context diagram. [1]

(*ii*) Draw the level 1 DFD and level 2 DFDs. [2]

(*iii*) Draw the structure chart. [2] 5

Or

(*a*) What do you mean by Widgets ? Briefly explain some important widgets using which you can develop a graphical user interface for any application. 4

(*b*) What do you understand by a metaphor in user interface design ? Why it is advantageous ? List some metaphors which can be used for user interface design. 4

4. (*a*) What do you mean by the term software re-engineering ? Discuss with a neat diagram. Why is it required ? Explain the different activities undertaken during re-engineering. 4

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6. (a) Why is it important for a software development organization to obtain ISO 9001 certification ? With the help of suitable examples, discuss the types of software organizations to which ISO 9001, 9002 and 9003 standards respectively are applicable. Discuss the merit of ISO 9000 and SEI CMM. 4

(b) What do you understand by performance testing ? Briefly explain the different kinds of performance testing that are usually performed on large software products ? 4

Or

(a) Schematically draw the architecture of a CASE environment and explain how the different tools are integrated. Discuss the role of the data dictionary in a CASE environment. 4

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(*b*) What is a Window Management System (WMS) ? Represent the main components of a WMS in a schematic diagram and briefly explain their roles. 4

Or

(*a*) What do you mean by integration testing ? Why it is required ? Explain the various approaches for integration testing with their advantages and disadvantages. 4

(*b*) What do you mean by coupling ? Why low coupling is desirable among the modules while developing a software product ? Explain the different types of coupling with suitable examples. 4

5. (*a*) What do you understand by software reliability ? Though the simple plot explain how the reliability of a software and hardware product changes over its lifetime. Why the two plots look so different ? 4

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(*b*) How coding standard is different from coding guideline ? Briefly discuss the guidelines for good coding style with suitable examples. 4

Or

(*a*) Explain using simple sentence what do you understand by the following reliability measures : 4

(*i*) A POFOD of 0.001

(*ii*) A ROCOF of 0.002

(*iii*) A availability of 0.998.

(*b*) What do you understand by software configuration management ? How is it carried out ? What problems would you face if you are developing several versions of the same product according to a client's request, and you are not using any configuration management tools ? 4

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(*b*) What is the difference between static and dynamic models in the context of object-oriented modeling of systems ? Identify the UML diagrams which provide these two models respectively. What is stereo-type in UML ? 4

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